Incident handler's journal

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Cybersecurity Lab Experience & Case Documentation

Date: 18/07/2025.	**Entry: 1**
Description	A phishing email contained malicious attachment was sent to U.S. health care clinic where a security incident took place on Tuesday at 9:00 am, where an employee downloaded the attachment triggering a ransomware attack (encrypting the orgs computer files). Attack was carried out by an organized group of unethical hackers. They left a ransom note demanding money in order to decrypt those files.
Tool(s) used	None
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? An organized group of unethical hackers who are known to target organizations in healthcare and transportation industries caused the incident. • **What** happened? A phishing email containing malicious files was sent to several employees. Once downloaded the file installed malware on employees computer, which triggered a ransomware attack that encrypted the organization's files • **When** did the incident occur? The incident occurred on Tuesday at 9:00 am • **Where** did the incident happen?

	The incident happened at a small U.S. healthcare clinic that specializes in delivering primary health-care services. • **Why** did the incident happen? The event occurred because of an employee, who opened the malicious email attachment containing malware. This allowed the attacker to gain access to the system and deploy ransomware, resulting in file encryption and business disruption.
Additional notes	Every company should provide regular cybersecurity awareness training to employees, staff must be educated about the common threats such as phishing, vishing, smishing attacks. These are most common among the cybercriminals. Teaching employees how to identify and report these threats can help prevent data breaches, reduce business disruption and protect company assets.

Date: 22/07/2025	**Entry: 2**
Description	A phishing email with malicious attachment was sent to finance service company, an employee had opened the malicious email and opened the attachment possibly containing malware
Tool(s) used	VirusTotal, SIEM (security information and event management)
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? An unknown threat actor (external attacker) initiated the phishing attempt. • **What** happened? A phishing email with grammatical errors and a suspicious attachment (bfsvc.exe) was sent to the HR department. The attachment was password-protected and, once opened, triggered a security alert. The hash of the file matched a known malicious signature. • **When** did the incident occur? The incident occurred on wednesday, 20th july 2022 at 9:30:14 AM • **Where** did the incident happen? At a financial services company, specifically targeting the HR department • **Why** did the incident happen? An employee at the company opened an email containing malicious attachment (malware) which resulted in a triggered alert
Additional notes	There's a mismatch between the sender's email address and sender's name, the body message and subject line contain grammatical errors, also after checking the attachment file's hash we discovered that the attachment has already been malicious.

Date: 23/07/2025	**Entry: 3**
Description	A mid-sized retail company faced a security incident. An unknown individual was able to gain unauthorized access to customer personal identifiable information (PII) and financial information.
Tool(s) used	Web server log
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? An unknown individual caused the incident, where he was able to access the personal identifiable information (PII) and financial information of customers. • **What** happened? The root cause was a vulnerability in the e-commerce web application. The attacker used a forced browsing technique by modifying the order number in the URL of the purchase confirmation page. This gave them access to customer financial and personal data. • **When** did the incident occur? - On December 22, 2022, at approximately 3:13 PM, an employee received an email from an external address claiming to have stolen customer data and demanding a $25,000 cryptocurrency payment. - On December 28, 2022, at 7:20 PM, the same employee received another email from the same sender, which included a sample of the stolen data and increased the ransom demand to $50,000. • **Where** did the incident happen? The incident occurred at mid-sized retail company's e-commerce website • **Why** did the incident happen?

	The incident occurred due to a vulnerability in the web application that allowed the attacker to exploit it through forced browsing, enabling access to unauthorized pages containing customer data.
Additional notes	The incident happened because the website didn't properly check who was allowed to view certain pages. The attacker took advantage of this and accessed many customer records. An earlier warning email was ignored, which gave the attacker more time. Checking the website logs helped find out what happened. This shows the need for regular security checks, employee training, and quick reporting of any suspicious activity.

Date: 24/07/2025	**Entry: 4**
Description	Configured and executed Suricata with a custom rule set to monitor simulated network traffic from a sample.pcap file. Analyzed generated alerts in fast.log and detailed logs in eve.json to understand rule effectiveness and incident context.
Tool(s) used	1. Suricata (IDS/IPS) 2. sample.pcap (packet capture) 3. custom.rules (rule file) 4. fast.log (alert output) 5. eve.json (standard event log)
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? A simulated attacker embedded in the packet capture file (sample.pcap), crafted to trigger custom Suricata rules.

	• **What** happened? Suricata detected traffic matching the signature criteria defined in the custom.rules file. An alert was triggered based on suspicious HTTP activity in the .pcap file. • **When** did the incident occur? During the lab exercise on 24/07/2025, at the time Suricata was executed with the custom rule against the sample.pcap file. • **Where** did the incident happen? Locally on the lab system using Suricata, analyzing traffic stored in the sample.pcap. • **Why** did the incident happen? The incident was intentionally triggered using test traffic to verify if the custom Suricata rule properly identifies specific traffic patterns or malicious payloads.
Additional notes	This lab exercise improved my understanding of rule logic, alert generation, and log analysis with Suricata.

Date: 25/07/2025	**Entry: 5**
Description	Used Splunk Cloud to investigate potential security issues on the Buttercup Games mail server. Uploaded and indexed log data from a provided .zip file and performed SPL-based queries to identify failed SSH login attempts to the root account. This helped simulate a real-world scenario of incident detection using SIEM tools.

Tool(s) used	1. Splunk Cloud (SIEM) 2. Search Processing Language (SPL) 3. tutorialdata.zip (log files) 4. Web-based Splunk interface
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? Potential unauthorized users attempting to log in to the mail server via SSH using the root account. • **What** happened? Multiple failed SSH login attempts were observed targeting the root user account on the mail server “mailsv”, indicating possible brute-force or unauthorized access attempts. • **When** did the incident occur? The events were discovered and analyzed on 25/07/2025 during the Splunk lab session. • **Where** did the incident happen? On the Buttercup Games mail server (host=mailsv) as captured in the /mailsv/secure.log file. • **Why** did the incident happen? The failed SSH login attempts may have been caused by someone trying to break into the system using the root account. It could also be part of a security test. Either way, failed root logins are a common sign of a possible attack and should be investigated.
Additional notes	This Splunk activity gave me hands-on experience using log data to detect security issues. I learned how to search for failed SSH login attempts using SPL, which helped me understand how attackers try to access systems. It felt like real-world analyst work and showed how useful Splunk is for identifying threats quickly.

Reflections/Notes: Record additional notes.

1. **Were there any specific activities that were challenging for you? Why or why not?**
   Yes, I found some of the networking-related tasks a bit challenging, especially when it came to analyzing network traffic and understanding certain packet-level details. Tools like Wireshark and tcpdump were new to me, and it took some time to get comfortable reading and interpreting network data. However, with practice in the labs and hands-on activities, I gradually improved and started to understand how network-based threats are detected and analyzed.

2. **Has your understanding of incident detection and response changed since taking this course?**
   Yes, this course helped me a lot. It deepened my understanding of incident detection and response. I now have a clearer understanding of concepts like IDS (Intrusion Detection Systems), including both NIDS (Network-based IDS) and HIDS (Host-based IDS), as well as IPS (Intrusion Prevention Systems).
   I also learned how important documentation, proper processes, and structured analysis are in incident response. These elements are essential to investigating and handling security incidents effectively.

3. **Was there a specific tool or concept that you enjoyed the most? Why?**
   Yes, I really enjoyed using tools like Suricata, Splunk, and OSINT platforms. Practicing in labs using Kali Linux, Wireshark, and tcpdump made the experience even more engaging. These tools gave me a real sense of what it's like to work as a cybersecurity analyst.
   The hands-on activities helped me understand how to detect and analyze threats, monitor network traffic, and respond like a real professional. It was exciting to apply the concepts in a simulated environment — it felt like I was actually working in a real-world SOC (Security Operations Center).